

Comprehensive Measure of Well-Being (HDI Predictor)

Project Description:

The **Human Development Index (HDI) Predictor** is a Machine Learning-based web application that predicts a country's Human Development Index (HDI) score using key socio-economic indicators such as **Life Expectancy**, **Expected Years of Schooling**, **Mean Years of Schooling**, and **Gross National Income (GNI) per Capita**. The application is developed using **Python**, **Flask**, **Pandas**, **NumPy**, **Scikit-learn**, **Matplotlib**, and **Seaborn**.

The system preprocesses the dataset, trains a Linear Regression model, and stores the trained model using Pickle. Users can enter the required values through a simple web interface, and the application predicts the HDI score along with its corresponding development category (**Very High**, **High**, **Medium**, or **Low**). An Analysis page provides visualizations such as heatmaps, scatter plots, pair plots, strip plots, and distribution plots to help users understand the relationships between different development indicators.

The project provides a user-friendly platform for students, researchers, policymakers, and development organizations to analyze human development indicators and understand the factors influencing a country's overall development.

Scenarios

Scenario 1: Predicting Highly Developed Countries

A policymaker enters high values for life expectancy, education, and GNI per capita. The application predicts a **Very High HDI** score, indicating that the country has excellent healthcare, education, and living standards.

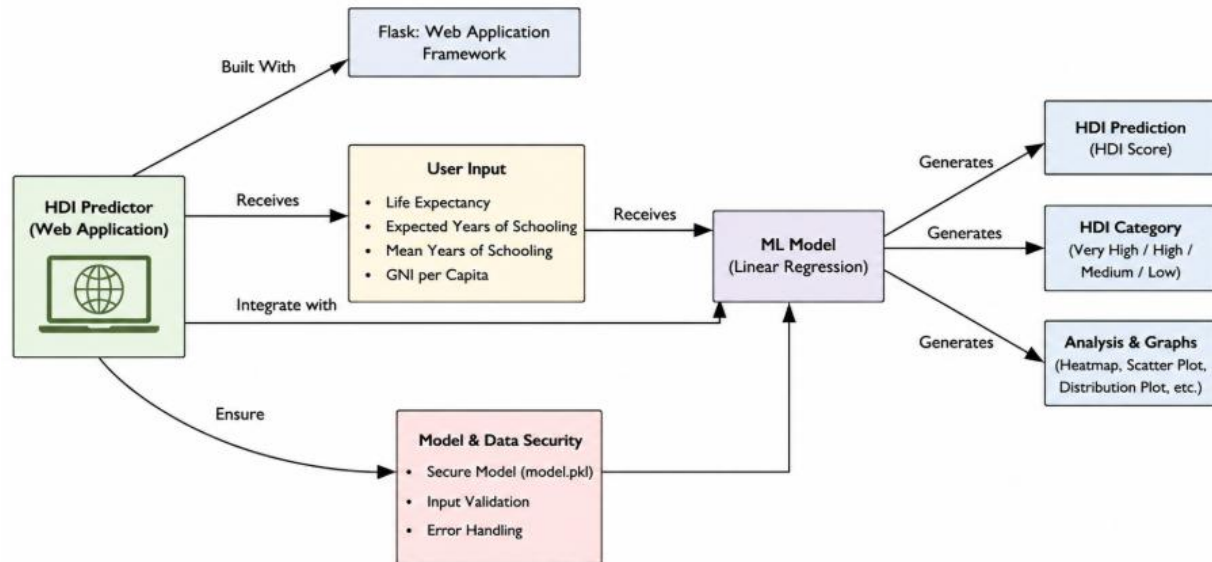
Scenario 2: Evaluating Developing Countries

A researcher provides moderate values for life expectancy, years of schooling, and income. The model predicts a **Medium HDI** score, helping identify areas where improvements in education, healthcare, or income could enhance overall human development.

Scenario 3: Assessing Low Human Development

A student analyzes a country with low life expectancy, limited educational opportunities, and low GNI per capita. The application predicts a **Low HDI** score, highlighting the need for policy interventions and investment in health, education, and economic development.

Technical Architecture:



The **Human Development Index (HDI) Predictor** follows a modular machine learning architecture to predict the Human Development Index of a country using key socio-economic indicators. The user enters values such as **Life Expectancy**, **Expected Years of Schooling**, **Mean Years of Schooling**, and **Gross National Income (GNI) per Capita** through the web interface. The **Flask** application receives and validates the input, then passes it to the trained **Linear Regression** model (`model.pkl`) for prediction. The model generates the **HDI score**, classifies it into **Very High**, **High**, **Medium**, or **Low**, and displays the prediction along with data visualizations such as **Heatmap**, **Scatter Plot**, **Distribution Plot**, **Pair Plot**, and **Strip Plot**. The application provides a simple, interactive, and efficient platform for HDI prediction and analysis.

Pre-requisites

- **Python Programming:** Basic knowledge of Python 3.x
- **Flask Framework:** Understanding of Flask web development
- **Machine Learning:** Basic concepts of Linear Regression and Scikit-learn
- **Data Analysis:** Familiarity with Pandas and NumPy
- **Data Visualization:** Knowledge of Matplotlib and Seaborn
- **Frontend Development:** Basic HTML and CSS
- **Version Control:** Basic Git and GitHub knowledge
- **Development Environment:** Visual Studio Code or Jupyter Notebook
- **Dataset:** Human Development Index (HDI) dataset in CSV format

Project Workflow:

Milestone 1: Model Selection and Architecture

- **Activity 1.1:** Collect the Human Development Index (HDI) dataset from Kaggle and understand its features.
 - **Activity 1.2:** Research and select the **Linear Regression** algorithm for HDI score prediction.
 - **Activity 1.3:** Design the project architecture, including the frontend, Flask backend, machine learning model, and dataset.
 - **Activity 1.4:** Set up the development environment by installing Python, Flask, Pandas, NumPy, Scikit-learn, Matplotlib, and Seaborn.
-

Milestone 2: Core Functionalities Development

- **Activity 2.1:** Perform data preprocessing by cleaning the dataset, handling missing values, and selecting important features.
 - **Activity 2.2:** Train the Linear Regression model, evaluate its performance using R^2 Score, MAE, and MSE, and save the trained model using Pickle.
-

Milestone 3: App.py Development

- **Activity 3.1:** Develop the Flask application (`app.py`) to manage routing, receive user input, load the trained model, and generate HDI predictions.
-

Milestone 4: Frontend Development

- **Activity 4.1:** Design responsive web pages using HTML, CSS, and Bootstrap for Home, Prediction, Analysis, and About pages.
 - **Activity 4.2:** Integrate the Flask backend with the frontend to display prediction results and analytical visualizations dynamically.
-

Milestone 5: Testing and Deployment

- **Activity 5.1:** Test the application to ensure accurate predictions, proper navigation, and correct visualization of graphs.
 - **Activity 5.2:** Prepare the project for deployment and upload the complete source code and documentation to GitHub.
-

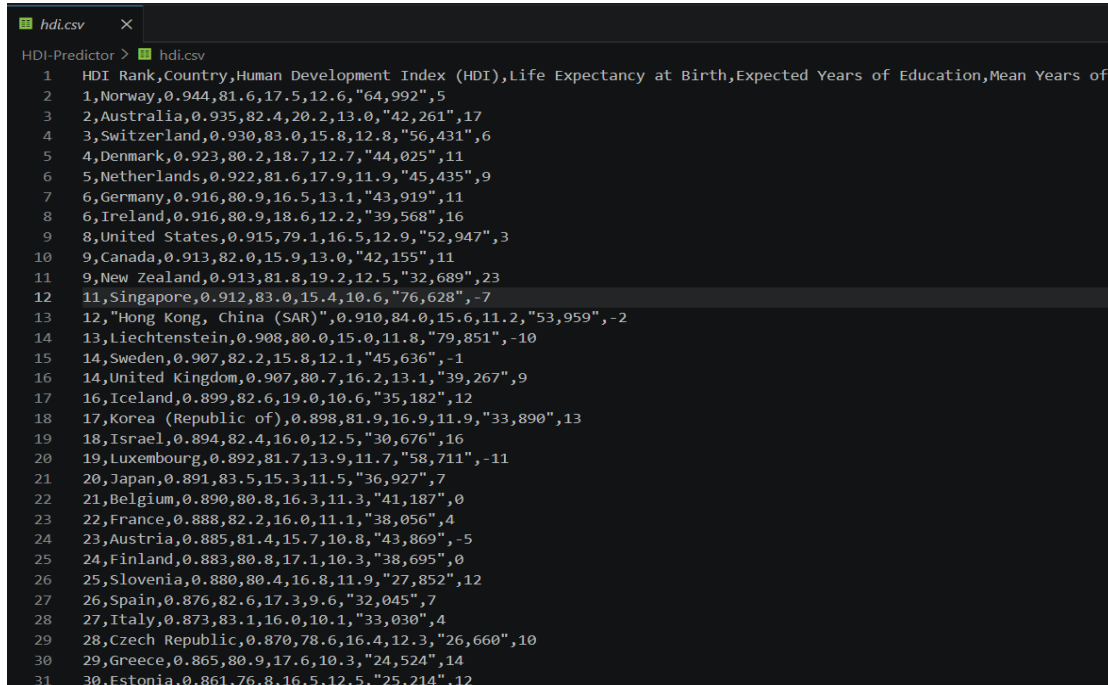
Milestone 6: Conclusion

- **Activity 6.1:** Successfully develop and deploy the HDI Predictor capable of predicting HDI scores and categories.

Milestone 1: Model Selection and Architecture

Activity 1.1: Dataset Collection and Understanding

The Human Development Index (HDI) dataset was collected from Kaggle. The dataset contains important socio-economic indicators such as **Life Expectancy**, **Expected Years of Schooling**, **Mean Years of Schooling**, **Gross National Income (GNI) per Capita**, and the corresponding **HDI Score**. The dataset was analyzed to understand the relationship between these indicators and human development.

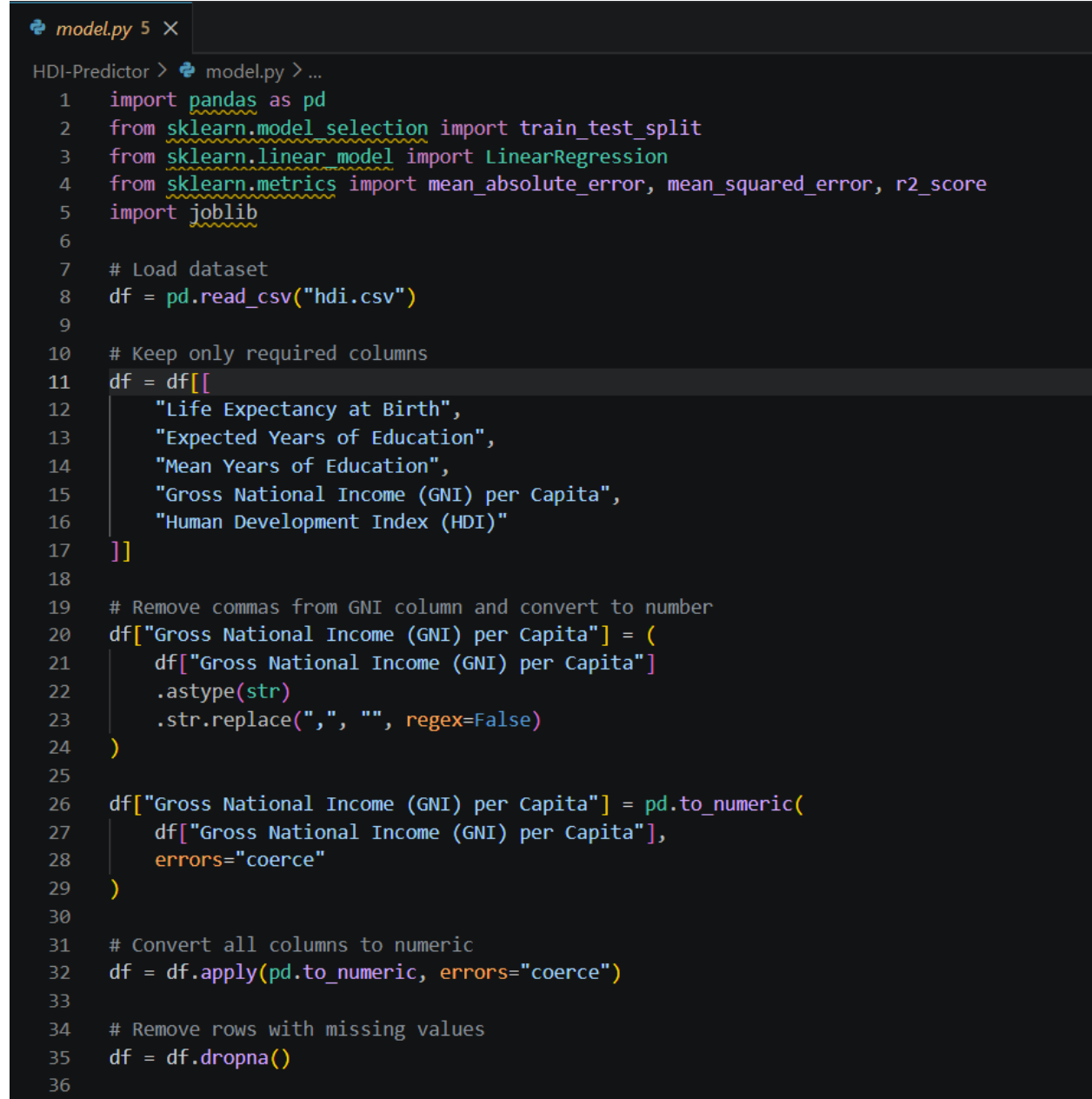


	HDI Rank	Country	Human Development Index (HDI)	Life Expectancy at Birth	Expected Years of Education	Mean Years of Schooling	Gross National Income (GNI) per Capita	HDI Score
1	1	Norway	0.944	81.6	17.5	12.6	"64,992"	5
2	2	Australia	0.935	82.4	20.2	13.0	"42,261"	17
3	3	Switzerland	0.930	83.0	15.8	12.8	"56,431"	6
4	4	Denmark	0.923	80.2	18.7	12.7	"44,025"	11
5	5	Netherlands	0.922	81.6	17.9	11.9	"45,435"	9
6	6	Germany	0.916	80.9	16.5	13.1	"43,919"	11
7	6	Ireland	0.916	80.9	18.6	12.2	"39,568"	16
8	8	United States	0.915	79.1	16.5	12.9	"52,947"	3
9	9	Canada	0.913	82.0	15.9	13.0	"42,155"	11
10	9	New Zealand	0.913	81.8	19.2	12.5	"32,689"	23
11	11	Singapore	0.912	83.0	15.4	10.6	"76,628"	-7
12	12	Hong Kong, China (SAR)	0.910	84.0	15.6	11.2	"53,959"	-2
13	13	Liechtenstein	0.908	80.0	15.0	11.8	"79,851"	-10
14	14	Sweden	0.907	82.2	15.8	12.1	"45,636"	-1
15	14	United Kingdom	0.907	80.7	16.2	13.1	"39,267"	9
16	16	Iceland	0.899	82.6	19.0	10.6	"35,182"	12
17	17	Korea (Republic of)	0.898	81.9	16.9	11.9	"33,890"	13
18	18	Israel	0.894	82.4	16.0	12.5	"30,676"	16
19	19	Luxembourg	0.892	81.7	13.9	11.7	"58,711"	-11
20	20	Japan	0.891	83.5	15.3	11.5	"36,927"	7
21	21	Belgium	0.890	80.8	16.3	11.3	"41,187"	0
22	22	France	0.888	82.2	16.0	11.1	"38,056"	4
23	23	Austria	0.885	81.4	15.7	10.8	"43,869"	-5
24	24	Finland	0.883	80.8	17.1	10.3	"38,695"	0
25	25	Slovenia	0.880	80.4	16.8	11.9	"27,852"	12
26	26	Spain	0.876	82.6	17.3	9.6	"32,045"	7
27	27	Italy	0.873	83.1	16.0	10.1	"33,030"	4
28	28	Czech Republic	0.870	78.6	16.4	12.3	"26,660"	10
29	29	Greece	0.865	80.9	17.6	10.3	"24,524"	14
30	30	Estonia	0.861	76.8	16.5	12.5	"25,214"	12

Figure 1.1: HDI Dataset Sample

Activity 1.2: Machine Learning Model Selection

Different machine learning algorithms were studied for predicting HDI scores. Since HDI is a continuous numerical value, **Linear Regression** was selected because it is simple, efficient, easy to interpret, and provides high prediction accuracy for this dataset. The trained model achieved an **R² Score of 0.9688**, indicating excellent prediction performance.



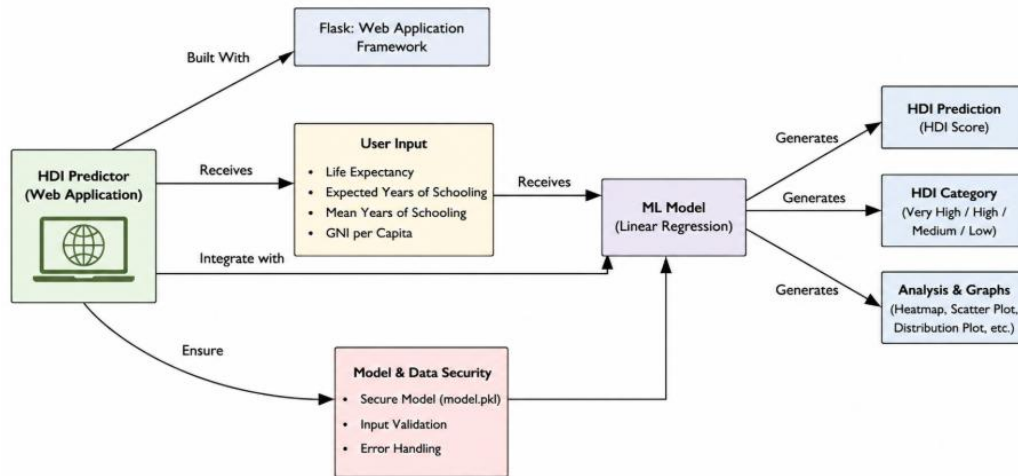
```

model.py 5 X
HDI-Predictor > model.py > ...
1  import pandas as pd
2  from sklearn.model_selection import train_test_split
3  from sklearn.linear_model import LinearRegression
4  from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
5  import joblib
6
7  # Load dataset
8  df = pd.read_csv("hdi.csv")
9
10 # Keep only required columns
11 df = df[[
12     "Life Expectancy at Birth",
13     "Expected Years of Education",
14     "Mean Years of Education",
15     "Gross National Income (GNI) per Capita",
16     "Human Development Index (HDI)"
17 ]]
18
19 # Remove commas from GNI column and convert to number
20 df["Gross National Income (GNI) per Capita"] = (
21     df["Gross National Income (GNI) per Capita"]
22     .astype(str)
23     .str.replace(",", "", regex=False)
24 )
25
26 df["Gross National Income (GNI) per Capita"] = pd.to_numeric(
27     df["Gross National Income (GNI) per Capita"],
28     errors="coerce"
29 )
30
31 # Convert all columns to numeric
32 df = df.apply(pd.to_numeric, errors="coerce")
33
34 # Remove rows with missing values
35 df = df.dropna()
36
  
```

Figure 1.2: Linear Regression Model Training

Activity 1.3: System Architecture Design

The project architecture follows a modular design consisting of a frontend developed using HTML and CSS, a Flask backend for request handling, a Linear Regression model for prediction, and an HDI dataset used for model training. The trained model is stored using Pickle and loaded during prediction. The system also generates graphical analysis using Matplotlib and Seaborn.



Activity 1.4: Set Up the Development Environment

The development environment was configured using Python 3.x and Visual Studio Code. Required libraries including Flask, Pandas, NumPy, Scikit-learn, Matplotlib, and Seaborn were installed using pip. Git and GitHub were used for version control and project management.

```

PS C:\Users\HP\javascript\practice\HDI-Predictor> pip install pandas numpy matplotlib seaborn scikit-learn joblib
The development environment was configured using Python 3.x and Visual Studio Code. Required libraries including Flask, Pandas, NumPy, Scikit-learn, Matplotlib, and Seaborn were installed using pip. Git and GitHub were used for version control and project management.
  
```

Figure 1.4: Development Environment Setup

Milestone 2: Core Functionalities Development

Activity 2.1: Data Preprocessing and Model Training

The HDI dataset was preprocessed before training the machine learning model. Missing values were handled, numerical values were converted into the correct format, and relevant features such as **Life Expectancy**, **Expected Years of Schooling**, **Mean Years of Schooling**, and **Gross National Income (GNI) per Capita** were selected. The dataset was then divided into training and testing sets, and a **Linear Regression** model was trained to predict the Human Development Index (HDI). The model achieved good accuracy with a high R^2 Score and low error values.

```

HDI-Predictor > model.py > ...
1 from sklearn.linear_model import LinearRegression
2 from sklearn.metrics import mean_absolute_error, mean_squared_error
3 import joblib
4
5 # Load dataset
6 df = pd.read_csv("hdi.csv")
7
8 # Keep only required columns
9 df = df[[
10     "Life Expectancy at Birth",
11     "Expected Years of Education",
12     "Mean Years of Education",
13     "Gross National Income (GNI) per Capita",
14     "Human Development Index (HDI)"
15 ]]
16
17 # Remove commas from GNI column and convert to number
18 df["Gross National Income (GNI) per Capita"] = (
19     df["Gross National Income (GNI) per Capita"]
20     .astype(str)
21     .str.replace(",", "", regex=False)
22 )
23
24
25

```

PROBLEMS 5 OUTPUT TERMINAL PORTS

▼ TERMINAL

```

PS C:\Users\HP\javascript\practice\HDI-Predictor> python model.py
File "<frozen importlib._bootstrap_external>", line 936, in exec_module
File "<frozen importlib._bootstrap_external>", line 1032, in get_code
File "<frozen importlib._bootstrap_external>", line 1130, in get_data
KeyboardInterrupt
● PS C:\Users\HP\javascript\practice\HDI-Predictor> python model.py
R2 Score: 0.9688323146199441
MAE: 0.019190034943047683
MSE: 0.000627970254503993
Model saved successfully!
○ PS C:\Users\HP\javascript\practice\HDI-Predictor>

```

Activity 2.2: HDI Prediction and Analysis Module

The trained model was integrated into the Flask application to provide real-time HDI predictions. Users enter socio-economic indicators through the web interface, and the application predicts the HDI score along with its development category (Very High, High, Medium, or Low). An Analysis page was also developed to visualize the dataset using Heatmaps, Scatter Plots, Distribution Plots, Pair Plots, and Strip Plots, helping users understand relationships among the indicators.

```

< index.html x
HDI-Predictor > templates > < index.html > html > body > footer
2  <html>
6  </head>
7  <body>
8  <div class="navbar">
9    <a href="/">Home</a>
10   <a href="/analysis">Analysis</a>
11   <a href="/about">About</a>
12 </div>
13 <div class="container">
14   <h1>Human Development Index (HDI) Predictor</h1>
15
16   <p>
17     Predict a country's Human Development Index using Machine Learning.
18     Enter the required indicators below to estimate the HDI score and development category.
19   </p>
20   <form action="/predict" method="POST">
21     <input type="number" step="any" name="life" placeholder="Life Expectancy at Birth" required>
22
23     <input type="number" step="any" name="expected" placeholder="Expected Years of Education" required>
24
25     <input type="number" step="any" name="mean" placeholder="Mean Years of Education" required>
26
27     <input type="number" step="any" name="gni" placeholder="GNI Per Capita" required>
28
29     <button type="submit">Predict</button>
30     <br><br>
31
32     <a href="/analysis">
33       <button type="button">View Data Analysis</button>
34     </a>
35
36   </form>
37

```

Figure 2.2: HDI Prediction Interface

```

<> result.html M X
HDI-Predictor > templates > <> result.html > html > body > div.container > p
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Prediction Result</title>
5      <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
6  </head>
7  <body>
8      <div class="navbar">
9          <a href="/">Home</a>
10         <a href="/analysis">Analysis</a>
11         <a href="/about">About</a>
12     </div>
13     <div class="container">
14         <div class="result-box">
15             <h2>Predicted HDI Score</h2>
16             <h1>{{ prediction }}</h1>
17             <h3>{{ category }}</h3>
18         </div>
19         <a href="/">
20             <button>Predict Again</button>
21         </a>
22         <hr>
23         <h3>Model Used</h3>
24         <p>Linear Regression</p>
25         <p>Features Used</p>
26         <ul>
27             <li>Life Expectancy</li>
28             <li>Expected Years of Education</li>
29             <li>Mean Years of Education</li>
30             <li>GNI Per Capita</li>
31         </ul>
32     </div>
33     <footer>

```

Figure 2.3: HDI Prediction Result

```

2  <html>
3  </html>
4
5  <body>
6
7  <div class="navbar">
8      <a href="/">Home</a>
9      <a href="/analysis">Analysis</a>
10     <a href="/about">About</a>
11
12 </div>
13
14 <div class="container">
15
16 <h1>HDI Data Analysis</h1>
17 <h2>Correlation Heatmap</h2>
18 
19
20 <h2>HDI Distribution</h2>
21 
22
23 <h2>HDI vs GNI</h2>
24 
25
26 <h2>Pair Plot</h2>
27 
28
29 <h2>Strip Plot</h2>
30 
31
32
33 <br><br>
34
35 <a href="/">
36 <button>Back to Predictor</button>
37 </a>

```

Figure 2.4: Data Analysis and Visualization Dashboard

Milestone 3: App.py Development

Activity 3.1: Flask Application Development

The `app.py` file was developed using the **Flask** framework to serve as the backend of the HDI Predictor application. It defines the application routes, receives user input from the web interface, validates the entered values, loads the trained machine learning model (`model.pkl`), predicts the Human Development Index (HDI) score, classifies the result into **Very High, High, Medium, or Low**, and displays the prediction through HTML templates. The application also provides navigation to the **Home, Analysis, and About** pages.

```
HDI-Predictor > app.py > about
4
5  app = Flask(__name__)
6
7  # Load trained model
8  model = joblib.load("model.pkl")
9
10 @app.route("/")
11 def home():
12     return render_template("index.html")
13 @app.route("/analysis")
14 def analysis():
15     return render_template("analysis.html")
16 @app.route("/about")
17 def about():
18     return render_template("about.html")
19 @app.route("/predict", methods=["POST"])
20 def predict():
21     life = float(request.form["life"])
22     expected = float(request.form["expected"])
23     mean = float(request.form["mean"])
24     gni = float(request.form["gni"])
25
26     features = np.array([[life, expected, mean, gni]])
27     prediction = model.predict(features)[0]
28
29     if prediction >= 0.800:
30         category = "Very High Human Development"
31     elif prediction >= 0.700:
32         category = "High Human Development"
33     elif prediction >= 0.550:
34         category = "Medium Human Development"
35     else:
36         category = "Low Human Development"
```

Figure 3.1: Flask Application (app.py) Implementation

Activity 3.1: Route Integration and Prediction

Flask routes were integrated with HTML templates to provide seamless interaction between the frontend and backend. User inputs are collected through the prediction form, processed by the Linear Regression model, and the predicted HDI score and category are displayed dynamically. The application ensures smooth navigation and accurate prediction results.

```
Debugger PIN: 112-803-467
PS C:\Users\HP\javascript\practice\HDI-Predictor> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 112-803-467
```

Figure 3.4: Flask Application Running Successfull

Milestone 4: Frontend Development

Milestone 4 focuses on developing a user-friendly and responsive web interface for the **Human Development Index (HDI) Predictor**. The frontend is designed using **HTML, CSS, and Bootstrap** to provide a clean, intuitive, and visually appealing user experience. The application allows users to enter HDI-related indicators, view prediction results, and analyze graphical visualizations through an interactive interface.

Activity 4.1: Designing and Developing the User Interface

Set Up the Base HTML Structure

- Develop the main HTML pages including **Home, Prediction, Analysis, and About** using semantic HTML elements.
- Design a responsive navigation bar to enable easy navigation between different pages.
- Create a prediction form for entering **Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and Gross National Income (GNI) per Capita**.

Design a Responsive Layout Using CSS

- Develop a CSS stylesheet (`static/style.css`) to create a clean and responsive interface.
- Use **Flexbox, CSS Grid, and Bootstrap** components to organize page content efficiently.
- Add media queries to ensure compatibility across desktop, tablet, and mobile devices.
- Apply consistent colors, typography, buttons, and card layouts to improve user experience.

Create Separate UI Components

- **Prediction Form:** Allows users to enter HDI-related input values.
 - **Result Card:** Displays the predicted HDI score and development category.
 - **Analysis Dashboard:** Displays Heatmap, Scatter Plot, Distribution Plot, Pair Plot, and Strip Plot.
 - **About Section:** Provides project details, technologies used, and workflow information.
-

Activity 4.2: Integrating the Frontend with Flask

Handle User Input

- Create a form that collects user inputs and submits them to the Flask backend.
- Validate user inputs before processing the prediction request.
- Display appropriate messages for invalid or missing inputs.

Display Prediction Results Dynamically

- Receive the predicted HDI score from the Flask backend.
- Display the predicted score and HDI category on the Result page.
- Navigate users seamlessly between Home, Prediction, Analysis, and About pages.

Render Analysis Visualizations

- Display analysis graphs generated using **Matplotlib** and **Seaborn**.
- Present visualizations such as **Heatmap**, **Scatter Plot**, **Distribution Plot**, **Pair Plot**, and **Strip Plot**.
- Ensure graphs are displayed clearly and responsively across different screen sizes.

Milestone 5: Deployment

Milestone 5 focuses on preparing the **Human Development Index (HDI) Predictor** for deployment and ensuring that the application runs successfully in a local environment. This includes configuring the development environment, installing the required dependencies, running the Flask application, testing all functionalities, and uploading the project to GitHub for version control and submission.

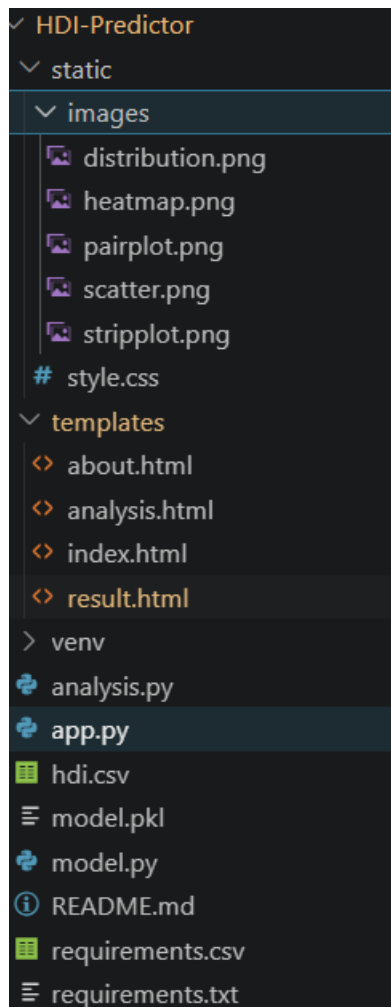
Activity 5.1: Preparing the Application for Local Deployment

Set Up the Development Environment

- Install **Python 3.x** and create a virtual environment for the project.
- Install all required dependencies using the `requirements.txt` file.
- Organize the project files, including the dataset (`hdi.csv`), trained model (`model.pkl`), templates, static files, and source code.
- Verify that all required Python libraries such as **Flask**, **Pandas**, **NumPy**, **Scikit-learn**, **Matplotlib**, and **Seaborn** are installed.

Configure the Project

- Ensure the trained model (`model.pkl`) is available in the project directory.
- Verify that the dataset and analysis images are correctly stored.
- Configure Flask routes and templates for smooth navigation between pages.



VS Code project structure

Activity 5.2: Running and Testing the Application

Run the Flask Application

- Start the Flask development server using the command:

```
python app.py
```

- Open a web browser and navigate to:

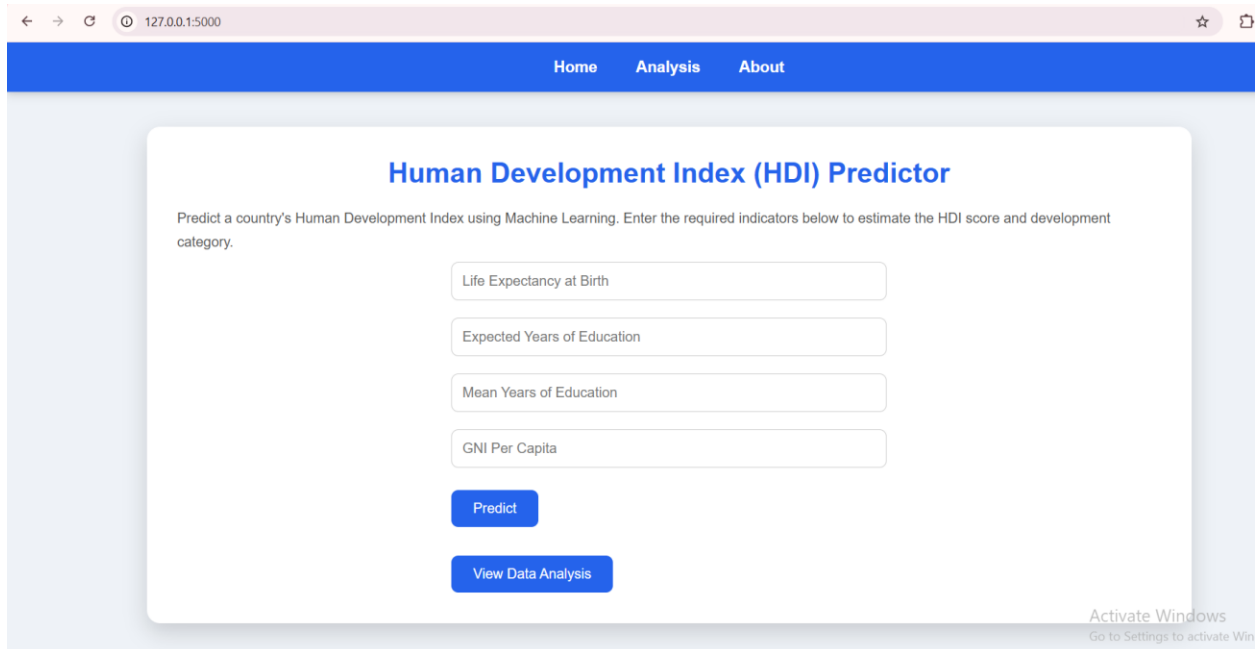
```
http://127.0.0.1:5000
```

- Verify that the Home, Prediction, Analysis, and About pages load successfully.

Test Application Features

- Enter valid input values and verify that the application predicts the HDI score correctly.
- Confirm that the HDI category (Very High, High, Medium, or Low) is displayed accurately.
- Check that all analysis graphs load correctly.
- Verify smooth navigation between all pages.

```
Debugger PIN: 112-803-467
PS C:\Users\HP\javascript\practice\HDI-Predictor> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 112-803-467
```



The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5000". The page has a blue navigation bar with links for "Home", "Analysis", and "About". The main content area is titled "Human Development Index (HDI) Predictor" and includes a subtitle: "Predict a country's Human Development Index using Machine Learning. Enter the required indicators below to estimate the HDI score and development category." Below this, there are four input fields labeled "Life Expectancy at Birth", "Expected Years of Education", "Mean Years of Education", and "GNI Per Capita". At the bottom of the form, there are two blue buttons: "Predict" and "View Data Analysis". In the bottom right corner of the browser window, there is a watermark that says "Activate Windows Go to Settings to activate Win".

Activity 5.3: Version Control and Project Submission

Upload the Project to GitHub

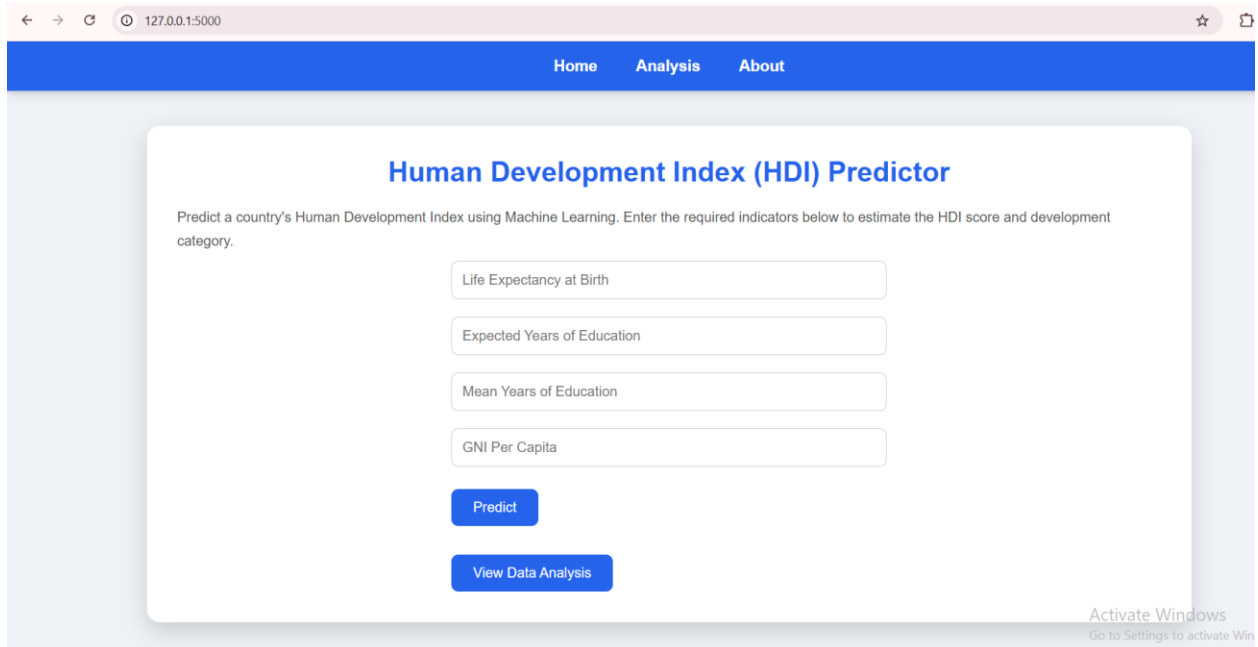
- Create a GitHub repository for the project.
- Upload the complete source code, documentation, dataset, templates, static files, and requirements file.
- Maintain a clear project folder structure with appropriate documentation.
- Verify that the repository is publicly accessible for evaluation

Exploring the Website's Web Pages

Home Page

Description

The **Home Page** serves as the main entry point of the Human Development Index (HDI) Predictor application. It introduces users to the purpose of the project and provides a simple, user-friendly interface for navigating through the application. The page contains a responsive navigation bar with links to the **Home**, **Analysis**, and **About** pages. It also includes a project overview explaining how machine learning is used to predict the Human Development Index based on socio-economic indicators. The clean layout, responsive design, and attractive styling ensure a smooth user experience across desktop and mobile devices.



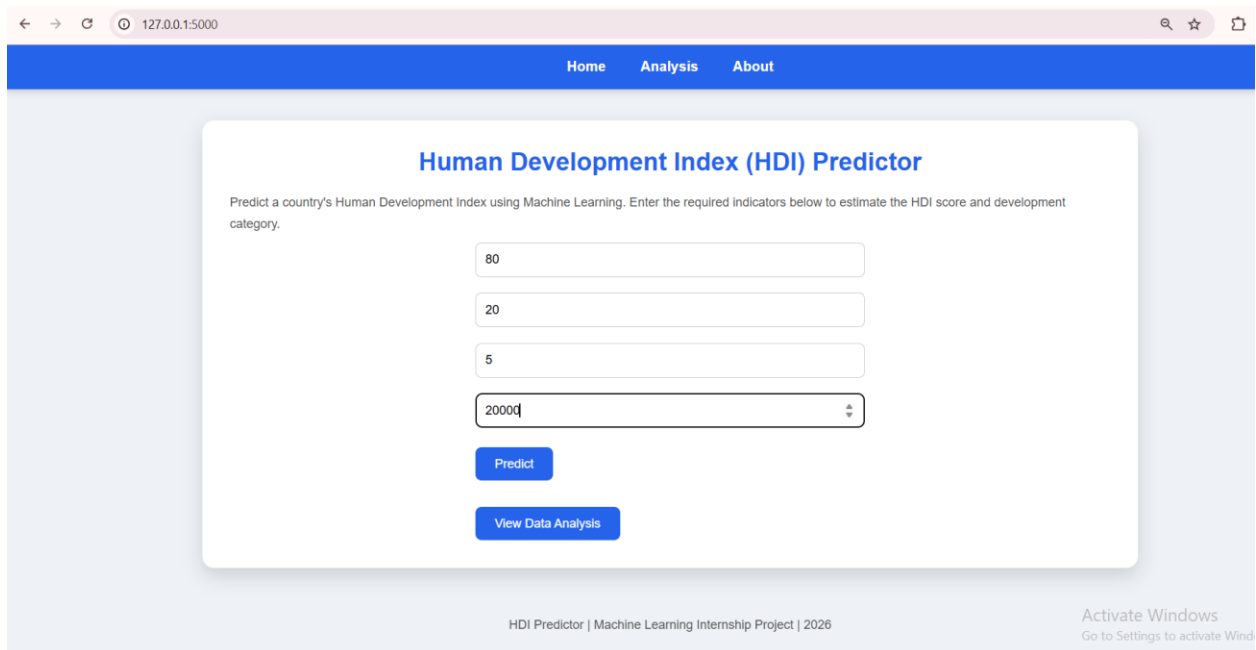
The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5000". The website has a blue navigation bar with links for "Home", "Analysis", and "About". The main content area is titled "Human Development Index (HDI) Predictor" and includes a description: "Predict a country's Human Development Index using Machine Learning. Enter the required indicators below to estimate the HDI score and development category." Below this, there are four input fields for "Life Expectancy at Birth", "Expected Years of Education", "Mean Years of Education", and "GNI Per Capita". There are two buttons: a blue "Predict" button and a blue "View Data Analysis" button. A watermark "Activate Windows" is visible in the bottom right corner.

Prediction Page Description

The **Prediction Page** enables users to predict the Human Development Index by entering the required socio-economic indicators. The form collects the following information:

- Life Expectancy at Birth
- Expected Years of Schooling
- Mean Years of Schooling
- Gross National Income (GNI) per Capita

After entering valid values, users click the **Predict** button to submit the data. The Flask backend validates the input, loads the trained Linear Regression model, and processes the information to generate an accurate HDI prediction. The page is designed with responsive input fields and buttons to provide an intuitive and seamless user experience.



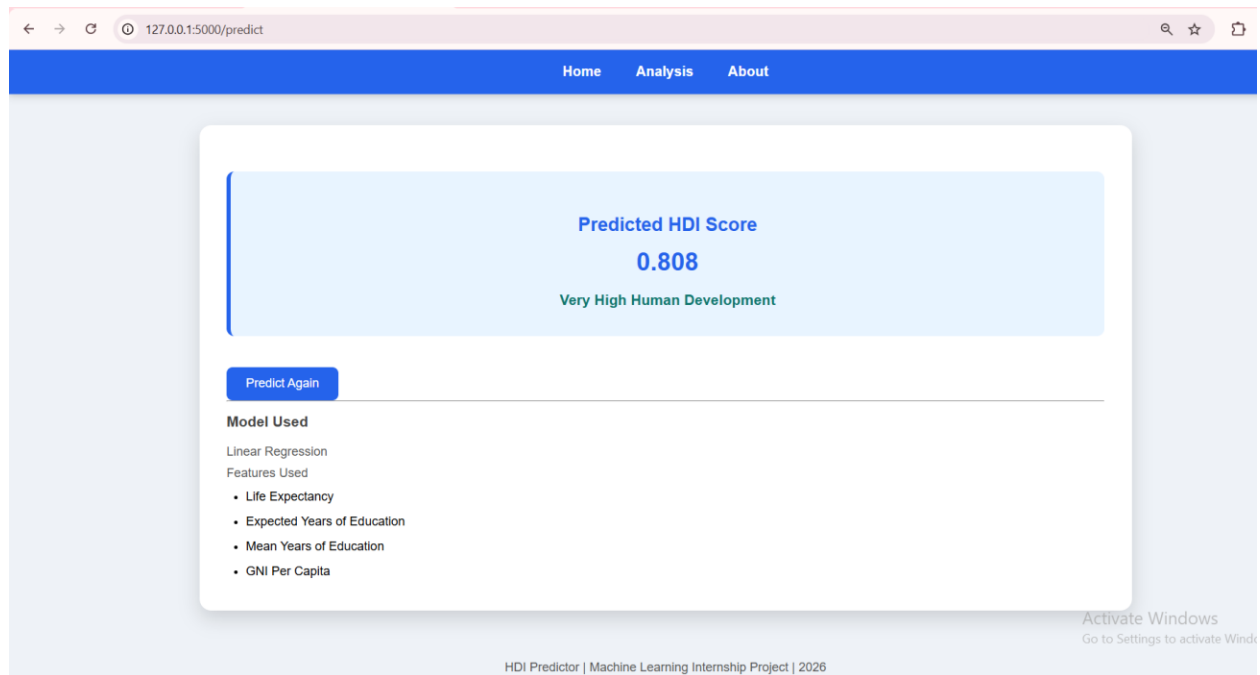
The screenshot shows a web browser window with the URL 127.0.0.1:5000. The application has a blue navigation bar with links for Home, Analysis, and About. The main content area is titled "Human Development Index (HDI) Predictor" and includes a subtitle: "Predict a country's Human Development Index using Machine Learning. Enter the required indicators below to estimate the HDI score and development category." Below this, there are four input fields: "Life Expectancy at Birth" (value: 80), "Expected Years of Schooling" (value: 20), "Mean Years of Schooling" (value: 5), and "Gross National Income (GNI) per Capita" (value: 20000). There are two buttons: "Predict" and "View Data Analysis". At the bottom, there is a footer with the text "HDI Predictor | Machine Learning Internship Project | 2026" and a Windows activation notice.

Prediction Result Page Description

The **Prediction Result Page** displays the output generated by the trained machine learning model. After processing the user inputs, the application predicts the **Human Development Index (HDI) Score** and classifies it into one of the four development categories:

- Very High Human Development
- High Human Development
- Medium Human Development
- Low Human Development

The predicted score is displayed clearly along with the corresponding category, allowing users to easily understand the development level based on the entered indicators. The result page is designed to present information in a simple and organized format.



Analysis Page Description

The **Analysis Page** provides visual insights into the HDI dataset through various statistical graphs generated using **Matplotlib** and **Seaborn**. These visualizations help users understand the relationship between different socio-economic indicators and the Human Development Index.

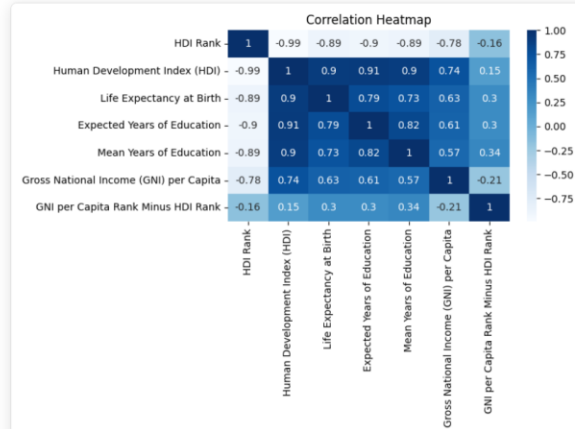
The page includes:

- **Correlation Heatmap** – Shows the correlation between numerical features.
- **Scatter Plot** – Displays the relationship between Gross National Income and HDI.
- **Distribution Plot** – Illustrates the distribution of HDI values.
- **Pair Plot** – Compares relationships among multiple features.
- **Strip Plot** – Displays the spread of HDI values.

These visualizations assist users in analyzing trends, identifying patterns, and understanding the factors that influence human development.

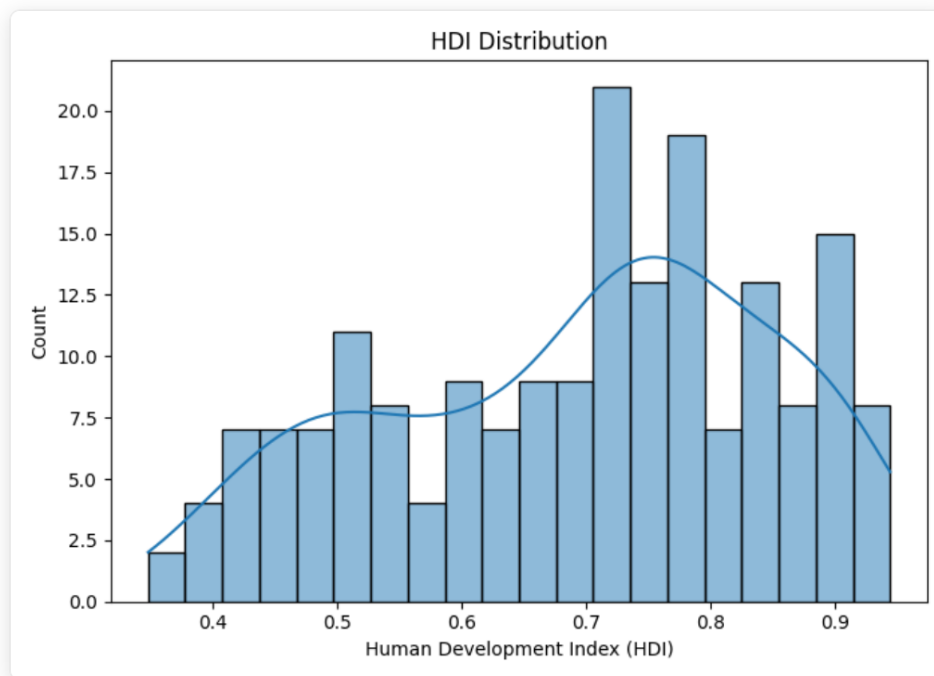
HDI Data Analysis

Correlation Heatmap



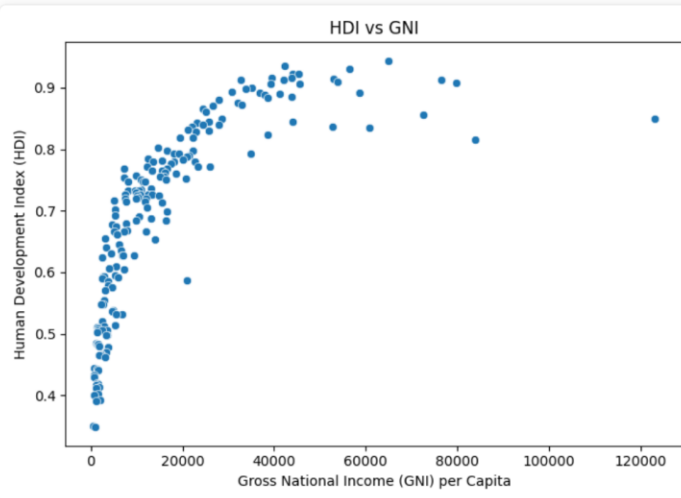
Activate Windows
Go to Settings to activate Windows

HDI Distribution



Activate Windows
Go to Settings to activate Windows

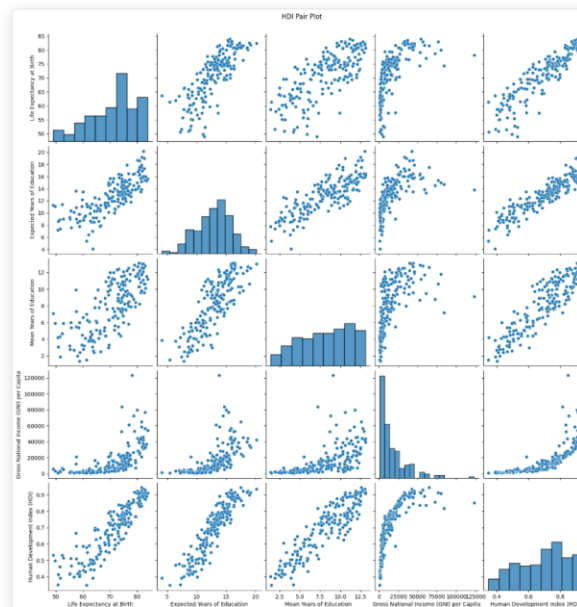
HDI vs GNI



Activate Windows

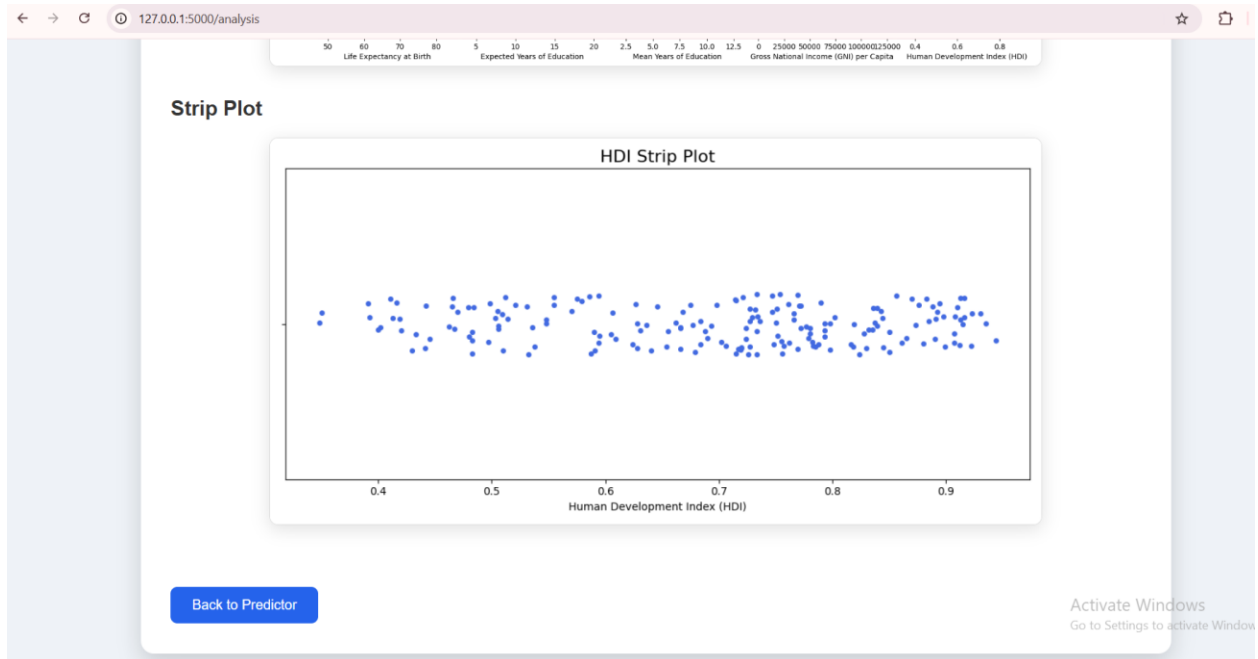
Go to Settings to activate Windows

Pair Plot



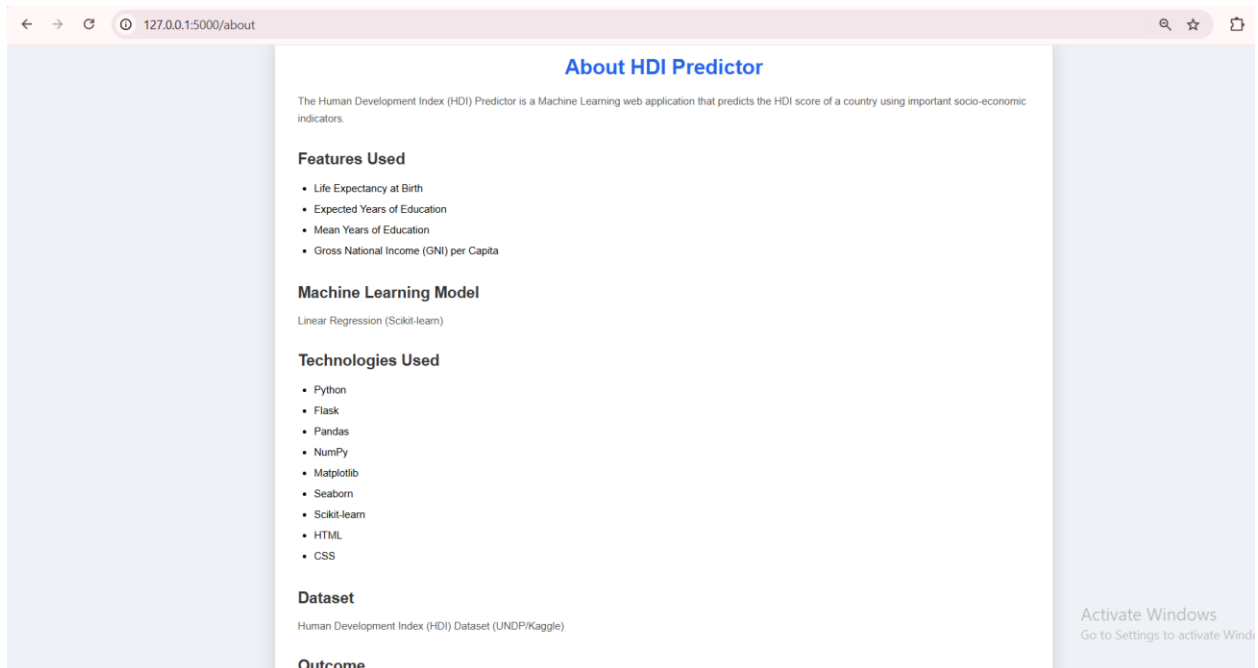
Activate Windows

Go to Settings to activate Windows



About Page Description

The **About Page** provides detailed information about the Human Development Index (HDI) Predictor project. It explains the project's objective, technologies used, machine learning model, and dataset information. The page also describes the overall workflow of the application, from data preprocessing and model training to prediction and visualization. This section helps users understand the implementation and purpose of the project while providing a brief overview of the tools and technologies used during development.



The image shows the "About HDI Predictor" page. The title "About HDI Predictor" is centered at the top. Below the title, there is a paragraph describing the project: "The Human Development Index (HDI) Predictor is a Machine Learning web application that predicts the HDI score of a country using important socio-economic indicators." The page is organized into sections with bullet points:

- Features Used**
 - Life Expectancy at Birth
 - Expected Years of Education
 - Mean Years of Education
 - Gross National Income (GNI) per Capita
- Machine Learning Model**
 - Linear Regression (Scikit-learn)
- Technologies Used**
 - Python
 - Flask
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-learn
 - HTML
 - CSS
- Dataset**
 - Human Development Index (HDI) Dataset (UNDP/Kaggle)
- Outcome**

At the bottom right, there is a watermark that says "Activate Windows Go to Settings to activate Windows".

Conclusion

The **Human Development Index (HDI) Predictor** is a machine learning-based web application developed to predict a country's HDI score using key socio-economic indicators such as **Life Expectancy**, **Expected Years of Schooling**, **Mean Years of Schooling**, and **Gross National Income (GNI) per Capita**. The project successfully integrates **Python**, **Flask**, **Scikit-learn**, **Pandas**, **NumPy**, **Matplotlib**, and **Seaborn** to provide accurate predictions along with meaningful data visualizations.

The application offers a simple and user-friendly interface that enables users to enter input values, view predicted HDI scores and categories, and analyze relationships between development indicators through interactive graphs. The trained **Linear Regression** model achieved high prediction accuracy, demonstrating the effectiveness of machine learning in analyzing human development.

Overall, the project provides a practical solution for students, researchers, policymakers, and development organizations to better understand and evaluate human development. In the future, the application can be enhanced by incorporating advanced machine learning models, cloud deployment, user authentication, country-wise comparisons, and real-time data integration to further improve its functionality and scalability.